

Notes Week 29

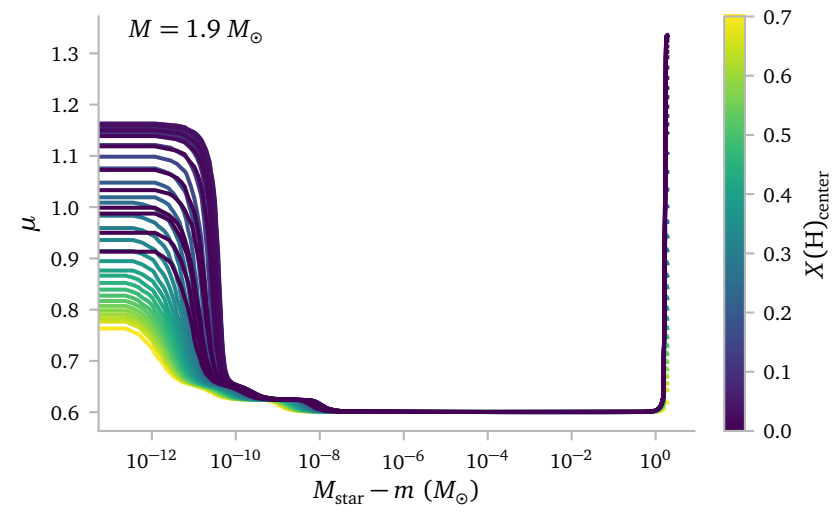
a. Thermohaline Mixing

I simulate a bunch of MS stars with different initial masses. I save profiles depending on the central hydrogen fraction to get an even coverage of the μ profiles throughout the MS evolution of the stars. This is important because the μ profiles change quite a lot throughout the MS evolution. We obviously know the age of the MS star when it accretes mass and thus we can make a pretty good assumption about the mixing.

The problem is then that we could possibly observe the MS stars billions of years after the mass has been accreted. At this point the μ profiles could change and it becomes uncertain.

However, since the abundances are quite uncertain anyway, I think it is okay to just take the μ profile at the time of accretion.

To start, let me plot the μ profile from the envelope to the center.

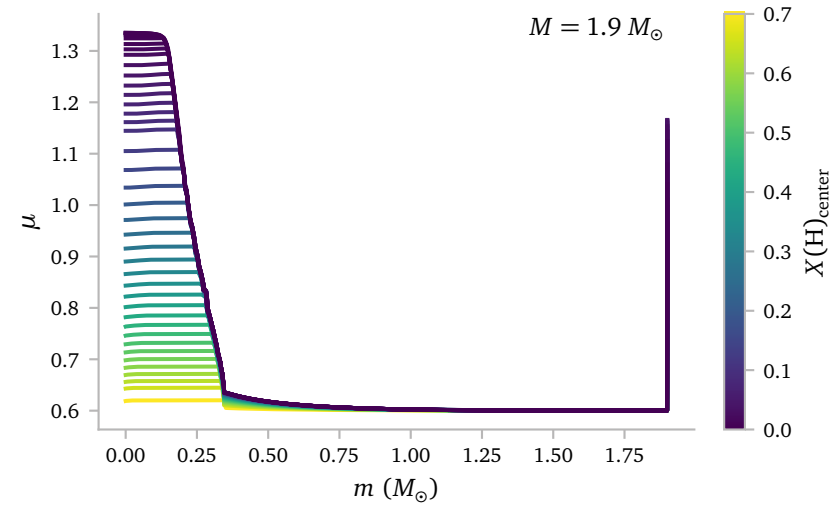


Here, we quite clearly see that outer layers of the envelope, which have very low density, are not fully ionized. This obviously causes the mean molecular weight to increase.

However, I think this is an unimportant effect. Accreted material will just the envelope layer further outwards, and the previously unionized material will become ionized.

The much more interesting part is close to the core.

Let's look at the profile going from the core outwards.

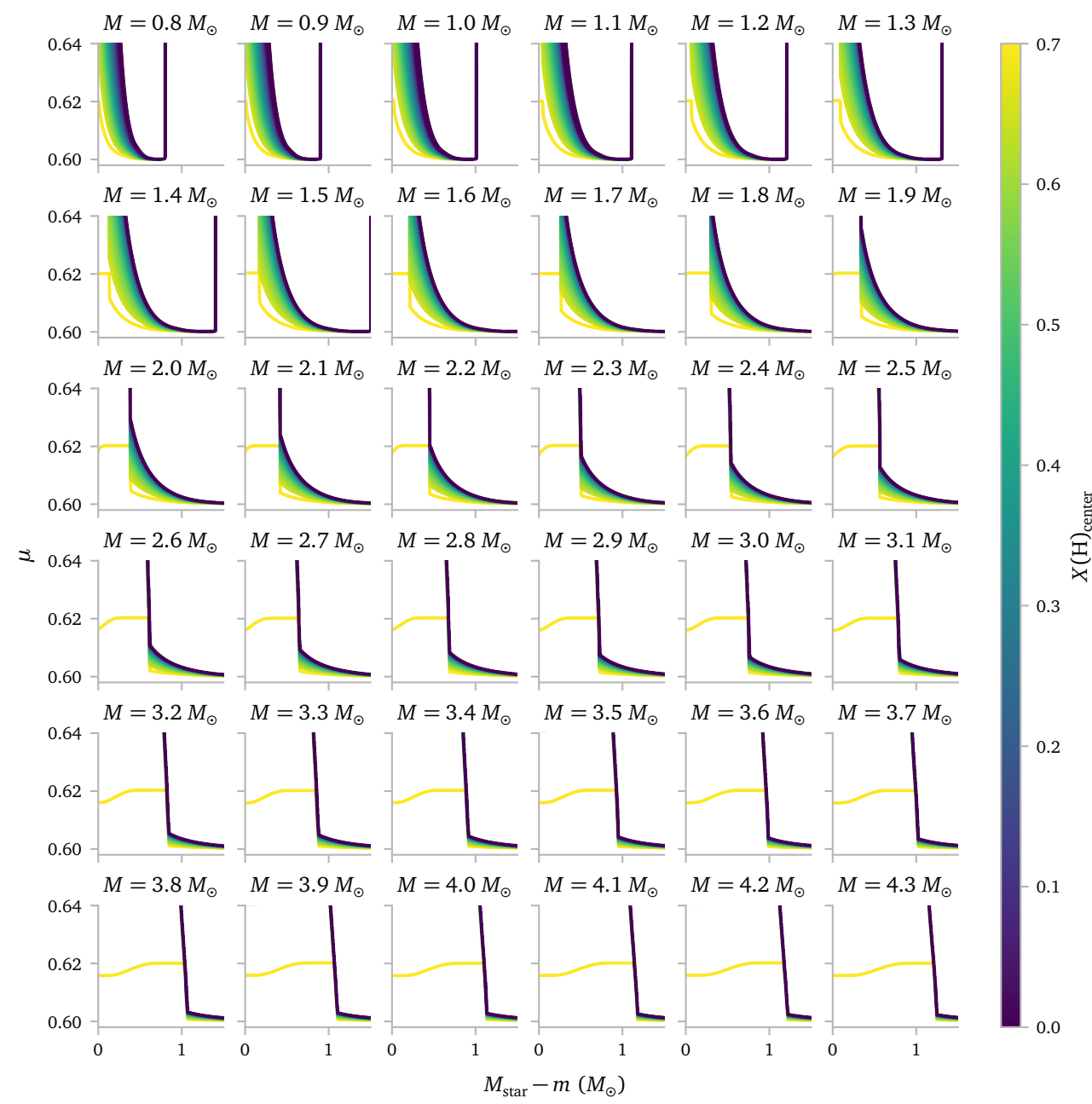


For this particular mass, we see a nearly perfectly diagonal line from the mean molecular weight in the core to the mean mean molecular weight in the envelope.

The diagonal line does not shift with the changing core composition, only the final height changes.

However, just before the "diagonal" line starts, the μ profile already starts to rise a bit. I think this is pretty interesting, as this seems to change quite a bit depending on the core composition, and could thus be important for the moment of accretion.

Below I plot all μ profiles for all models, zooming in on this lower portion.



Clearly, not only the mass position of the rise in μ changes with mass, but also the slow rise before the diagonal line.

However, I have no intuition for the mean molecular weight of some strongly enriched material, so before I do a bunch of calculations, let me make a really simple estimate for μ .

NOTE: Simple derivation for myself to refresh how the mean molecular weight is calculated.

The mean molecular weight is defined as:

$$\mu = \frac{\rho}{nm_u}.$$

Here,

- μ is the mean molecular weight of a gas.
- ρ is the mass density of a gas.
- n is the number density of a gas.
- m_u is the atomic mass unit.

For one element, the mass density contributed is

$$\rho_i = X_i \rho.$$

The number density of nuclei is then simply

$$n_i^{\text{nuc}} = \frac{\rho_i}{A_i m_u}.$$

However, if we assume full ionization, we will have 1 nucleus + Z_i electrons.

The "real" number density is simply

$$n_i = (1 + Z_i) n_i^{\text{nuc}}.$$

Now, let me expand and sum over all elements:

$$\begin{aligned} n &= \sum_i (1 + Z_i) \frac{\rho X_i}{A_i m_u} \\ &= \frac{\rho}{m_u} \sum_i \frac{X_i (1 + Z_i)}{A_i} \\ \frac{nm_u}{\rho} &= \sum_i \frac{X_i (1 + Z_i)}{A_i} \\ \frac{1}{\mu} &= \sum_i \frac{X_i (1 + Z_i)}{A_i} \end{aligned}$$

Cool! I am *assuming* full ionization here, which I think is fine. I am pretty sure nearly the whole MS star is ionized anyway.

Now to compute the mean molecular weight of the accreted gas, we need

- X_i , the mass fraction of every element in the accreted gas. We have this.
- Z_i , the proton number of every accreted element. We also have this.
- A_i , the mass number of every element in the accreted gas. We need to make an *assumption* here.

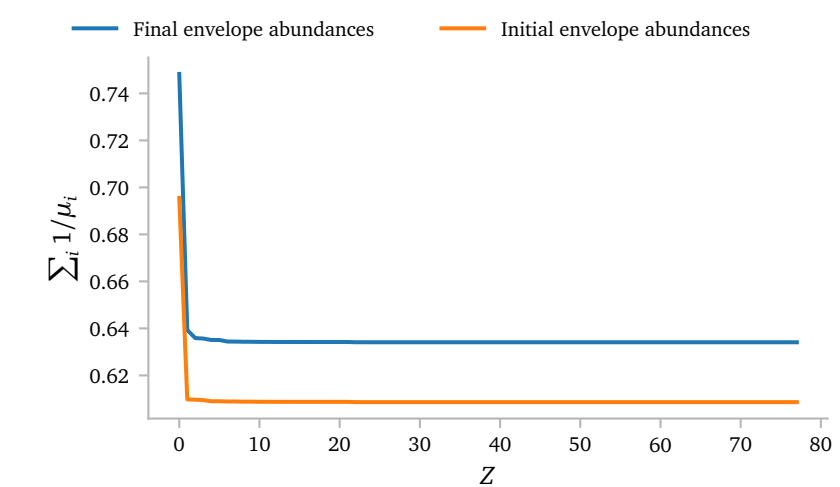
Especially for heavy elements, it is quite possible that the element has a few isotopes with different masses that are all stable and relatively abundant. In this case, the elemental mass should be the averaged mass of the isotopes.

I think it is fine to take the [standard atomic weight](#) in this case. The differences will be small anyway, especially for the heavy elements.

I will use the package `periodictable` written by [Kienzle et al. \(2026\)](#) based on data from [Meija et al. \(2016\)](#).

It is then quite straightforward to μ .

Below I plot the μ I calculated from my MESA model with the Monash Abundances.



There are a couple of important things here.

1. The first elements contribute by far the most to μ , which is to be expected due to their much larger abundances.
2. The μ computed with the final abundances is larger than for the initial envelope abundances.
3. μ computed from the initial envelope abundances is a bit larger than μ in the envelope of the MESA MS models, which have $\mu \approx 0.60$.

I am unsure if the differences in μ between the initial models is due to evolution after the MS, or due to differences between MESA and Monash abundances. But, in the end, the differences for the predicted abundances in the MS star would probably be small.

The idea is now to use this μ and mix the accreted mass, thereby changing μ as well, until the mixed μ is equal to the μ in the star.

References

Kienzle, P., Maranville, B. B., Cooper, J., et al. (2026) *python-periodictable*.
Meija, J., Coplen, T. B., Berglund, M., et al. (2016) *Atomic weights of the elements 2013 (IUPAC Technical Report)*. *Pure and Applied Chemistry*88, 265.